

Unit CAN

SKU:U085



Description

Unit CAN is a standalone Controller Area Network (CAN) transceiver unit, which can be used to build complex CAN communication networks. The built-in DC-DC isolation power chip can isolate noise and interference, preventing damage to sensitive circuits. The model of the isolated CAN transceiver is CA-IS3050G, which can provide differential reception and differential transmission capabilities. The bus can support up to 110 nodes, with a signal transmission rate of up to 1 Mbps. It features current limiting, overvoltage, ground loss protection (-40V to 40V), and thermal shutdown functions, preventing output short circuits, and complies with the ISO11898-2 standard technical specifications. CAN is an ISO international standard serial communication protocol. CAN belongs to the fieldbus category. It is a serial communication network that can effectively support distributed control or real-time control. Distributed control systems based on the CAN bus use a multi-master competition arbitration method for communication, characterized by multi-master operation, decentralized arbitration, and broadcast communication. It has obvious advantages in the following aspects: fast data communication speed between nodes in the network, low failure rate. When multiple nodes with the same priority communicate on the same line, congestion will be avoided. The communication distance can reach 10 km (rate less than 5 Kbps), and the rate can reach 1 Mbps (communication distance less than 40 m).

Features

- Built-in isolated DC-DC
- Power indicator light
- Signal rate up to 1 Mbps
- Protection functions: signal isolation, current limiting, overvoltage protection, thermal shutdown
- 2 LEGO-compatible holes
- HY2.0-4P interface

Includes

- 1 x Unit CAN
- 1 x HY2.0-4P Grove connection cable (20cm)
- 1 x HT3.96-4P
- 1 x 120R resistor

Applications

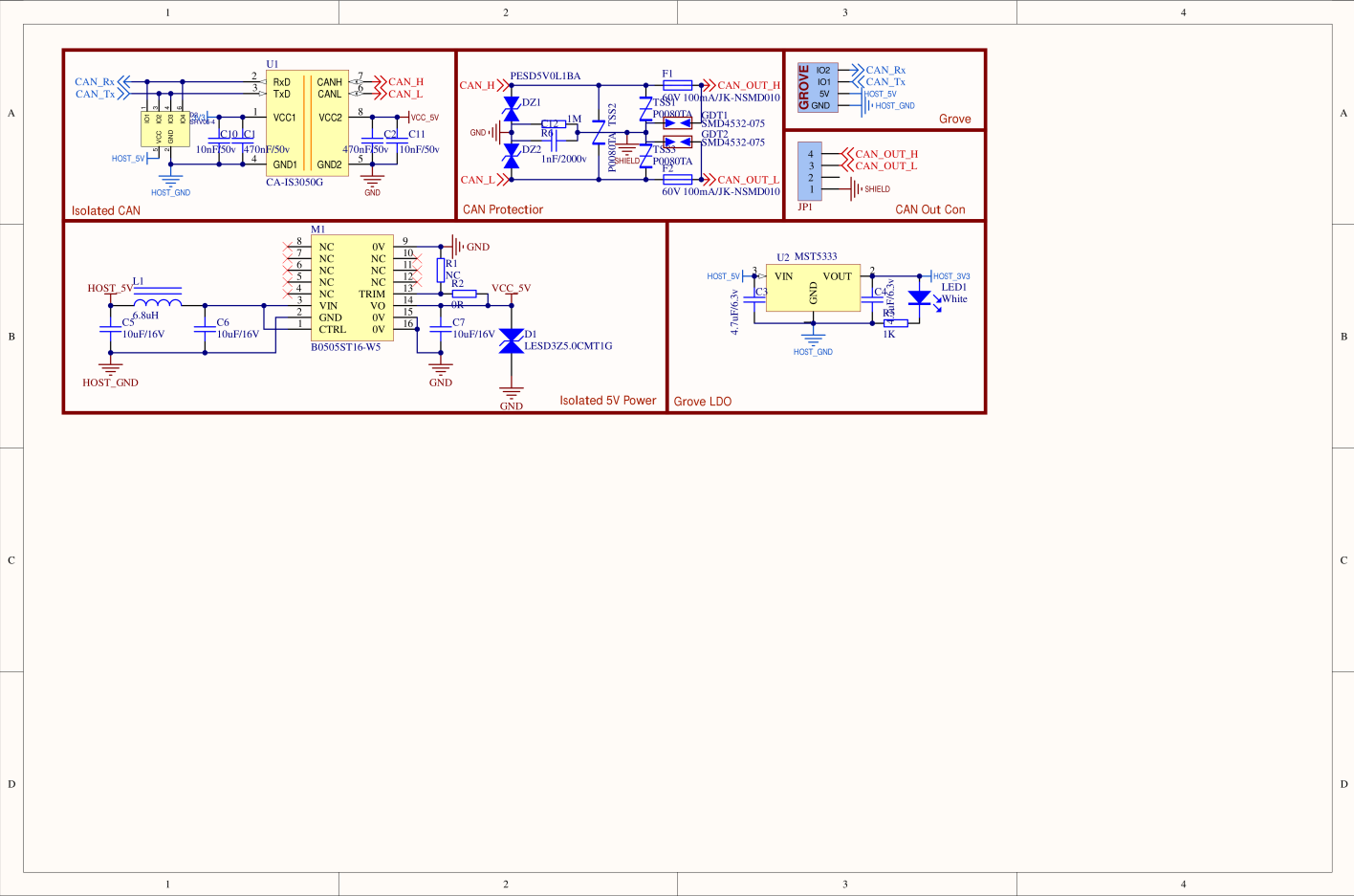
- CAN bus communication
- Industrial field control
- Security systems

Specifications

Specification	Parameter
Isolation Voltage	1000V
Maximum Rate	1 Mbps
Supported Nodes	110
Low Loop Delay	-150ns
Common Mode Voltage	-12V ~ 12V
Protection Functions	Current limiting, overvoltage, dominant timeout, thermal shutdown
Case Material	PC
Product Size	56.0 x 24.0 x 10.2mm
Product Weight	9.1g
Package Size	138.0 x 93.0 x 11.2mm
Gross Weight	17.1g

Schematics

- [Unit CAN Schematics PDF](#)



PinMap

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.C	GND	5V	CAN_TX	CAN_RX

Model Size

Unit CAN Model Size PDF

Easyloader	Download Link	Note
Unit CAN example with M5Core	download	/

Product Comparison

Product Compare	 	
	Unit Mini CAN	Unit CAN
CAN Transceiver	TJA1051T/3	CA-IS3050G
Power Isolation	Non-isolated	Isolated
Speed	1Mbps	1Mbps
Input Power	HY2.0-4P Interface DC 5V Power Supply HT3.96-4P Interface DC 9 ~ 24V Power Supply	HY2.0-4P Interface DC 5V Power Supply

Video

[CAN UNIT.mp4](#)